

Module – 3 (Contd..)

Uncertainty, type of uncertainty

Probabilistic Reasoning

joint distribution reasoning, Bayesian networks

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Uncertainty in the real-world

Reasons for uncertainty

- Randomness of events
(rolling dice, faulty oximeter/digital BP machine/sensor)
- Incomplete information
- Incorrect information
- Vagueness/imprecision in language
(tall, likely, small)
- Things may go wrong

Dealing with Uncertainty

Uncertainty refers to unpredictable situations where one does not have complete, reliable, or deterministic information about the environment, consequences of actions, or the outcomes of decisions.

Uncertainty in real-world applications

Robotics, autonomous vehicle, financial forecasts, etc. where it is not possible to know all variables precisely or to predict future events with certainty.

Types of uncertainty

1. State Uncertainty
2. Outcome Uncertainty
3. Model Uncertainty
4. Perceptual Uncertainty
5. Action Uncertainty
6. Temporal Uncertainty
7. Environmental Uncertainty

State Uncertainty

- When an agent does not have knowledge about the current state of its environment.
- Ex: A robot trying to navigate a room may not know the exact location of obstacles due to limited sensor range or faulty sensors.
- **Soln.:** Partial Observability; Sensing actions (as seen in conditional planning); Belief states (e.g., representing uncertainty about the state as a probability distribution over possible states).

Outcome Uncertainty

Refers to uncertainty in the results of actions. The agent knows what action it is taking but is unsure about the outcome due to probabilistic or unpredictable effects.

Ex: A robot that tries to pick up an object might succeed with a certain probability, but the object might slip or be too heavy, resulting in a failure.

Soln: Probabilistic models (e.g., Markov Decision Processes); Planning under stochastic environments.

Model Uncertainty

Arises when the agent does not have a complete or accurate model of the environment. The agent may not know all the dynamics, such as how certain actions affect the world or how certain variables interact.

Example: In a game, an agent might not know the complete set of rules or the opponent's strategy.

Soln: Machine learning models, Exploration strategies to gather more information.

Perceptual Uncertainty

- Caused by noisy or unreliable sensors and observations. The agent's perception of the world may be imperfect, leading to incorrect or ambiguous interpretation of the environment.
- Ex: A self-driving car might receive incomplete or noisy data from its sensors due to fog, rain, or equipment malfunction, making it difficult to detect obstacles accurately.
- **Soln.:** Sensor fusion (combining multiple sensor inputs to reduce uncertainty)

Action Uncertainty

- Occurs when the agent is uncertain about the effect of its own actions. The environment or internal system may not behave as expected due to unforeseen circumstances.
- Ex: An Expert system may prescribe a treatment, but it cannot be certain how a specific patient will respond to the medication.
- **Soln.:** Learning through feedback (Reinforcement Learning); Stochastic planning models.

Temporal Uncertainty

- Relates to the uncertainty about when certain events will happen or how long certain actions will take.
- Ex: An AI-controlled manufacturing process might be uncertain about the time it takes for a machine to finish a task, due to varying machine performance or external conditions.
- **Soln.:** Temporal probabilistic models (e.g., Temporal Bayesian networks); Real-time planning.

Environmental Uncertainty

- This occurs when the environment itself changes unpredictably, either due to external influences (e.g., due to other agents) or inherent dynamics.
- Ex: In an AI-based system for stock trading, the market can be influenced by many unpredictable factors such as political events or natural disasters/war etc.
- **Soln.:** Real-time data processing; Adaptive systems (models that can update themselves with new information).

Approaches to Handle Uncertainty in AI

- **Probabilistic Models**

- ✓ **Bayesian Networks**: Graphical models that represent conditional dependencies among variables.
- ✓ **Markov Decision Processes**: Used for decision-making in environments with probabilistic transitions between states.
- ✓ **Hidden Markov Models**: Used when a system's true state is hidden but can be inferred from observations.

- **Fuzzy Logic**

degrees of truth rather than strict true/false

- **Belief States**

probability distribution over possible states

- **Sensing and Active Learning**

- ✓ gather more information through sensing
- ✓ ask targeted questions to reduce uncertainty – active learning

- **Decision Theory and Expected Utility**

- Estimate the probable outcomes of an action and choose the action that provides the highest expected value.

- **Robust and Adaptive Systems**

- AI systems can be designed to be robust against certain types of uncertainty, meaning they can tolerate variations in data or outcomes without significant performance degradation.

- Adaptive systems learn and update their models in real-time to adjust to new information or changes in the environment, reducing model uncertainty.

Probabilistic Reasoning

- **Probabilistic reasoning** is a method of reasoning and decision-making under uncertainty, where the likelihood of different outcomes, events, or states is expressed as probabilities.
- Probabilistic reasoning allows AI systems to infer, predict, and act based on incomplete or uncertain information using probability theory.

Key Concepts in Probabilistic Reasoning

1. Probability: It is a numerical measure which indicates the likelihood of occurrence of an event. The value ranges from 0 (event will never occur) to 1 (event will surely occur).

2. Random Variables: These variables can take values that are outcomes of a random phenomenon.

- **Discrete random variables** take on a finite number of distinct values (e.g., True/False)
- **Continuous random variables** can take on any value in a range (e.g., temperature).

3. Probability Distribution:

- A probability distribution describes how likely different values of a random variable are. For example, a coin flip can be represented as a discrete probability distribution where the chances of getting heads or tails are each 0.5.
- **Joint probability distribution** gives the probability of combinations of values of multiple random variables.

4. Conditional Probability:

Conditional probability, denoted as $P(A|B)$, is the probability of event A occurring given that B is true. This helps one to know how knowledge of one event affects the likelihood of another event.

Ex: $P(\text{Rain}|\text{Cloudy})$ represents the probability that it will rain, given that it is cloudy.

5. Bayes' Theorem:

- It provides a way to update beliefs (probabilities) based on new evidence. It is used for inference and learning.

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

where $P(H|E)$ is the posterior probability of hypothesis H given evidence E , $P(E|H)$ is the likelihood of observing evidence E given H , $P(H)$ is the prior probability of H , and $P(E)$ is the overall probability of the evidence.

Example: Suppose an AI system is trying to diagnose whether a patient has a disease (H) based on a positive test result (E). Using Bayes' theorem, the AI system can update the probability of the patient having the disease after observing the test result.

6. Marginal Probability:

Marginal probability refers to the probability of a single event without considering the effects of other variables.

Ex: $P(\text{Rain})$ is the marginal probability that it will rain, regardless of whether it is cloudy or not.

Probabilistic Reasoning Models

1. Bayesian Networks (Belief Networks):

- A **Bayesian Network** is a directed acyclic graph where nodes represent random variables, and edges represent conditional dependencies between these variables. Each node has a conditional probability table (CPT) that quantifies the effect of its parent nodes.
- **Example:** In a medical diagnosis system, a Bayesian Network might include variables like "Symptoms," "Disease," and "Test Results," with edges representing the causal relationships among these factors.

2. Markov Networks (Markov Random Fields):

- A **Markov Network** is an undirected graph where nodes represent random variables, and edges represent dependencies between these variables. It is used in cases where relationships are not directional.
- Unlike Bayesian Networks, Markov Networks do not encode direct causal relationships but instead capture correlations between variables.

3. Hidden Markov Models (HMMs):

- An HMM is a statistical model that represents a system where the state is not directly observable (hidden) but can be inferred through observable outputs.
- HMMs are widely used in AI for sequential data, such as speech recognition or tracking, where the current state depends on both previous states and observations.
- Example: In speech recognition, the actual spoken words (hidden states) are inferred from the sound waves (observations) detected by the system.

4. Markov Decision Processes (MDPs):

- An MDP is a framework for modeling decision-making in environments where outcomes are partly random and partly under the control of the decision-maker.
- MDPs are used in reinforcement learning, where an agent learns to make optimal decisions by interacting with the environment.

Types of Probabilistic Reasoning

1.Diagnostic Reasoning (Inference from Effects to Causes):

This type of reasoning is used when we observe the effects and want to infer the possible causes.

Ex: Given the symptom of a fever, what is the probability that a person has cold?

Bayes' Theorem is often used for diagnostic reasoning.

2.Causal Reasoning (Inference from Causes to Effects):

In this case, we know the cause and want to predict its effects.

Ex: If a person has cold, what is the probability that he will have fever?

Causal reasoning is often represented using Bayesian Networks, which can model cause-and-effect relationships.

3.Prediction:

Probabilistic reasoning can be used to predict the likelihood of future events based on current evidence.

Ex: Given current weather conditions, what is the probability of rain tomorrow?

HMMs or Bayesian Networks can be used for predictive reasoning.

4.Decision-Making under Uncertainty:

Probabilistic reasoning can be a tool for decision-making where the outcomes of actions are uncertain.

Ex: In a self-driving car, the onboard AI system needs to decide whether to change lanes based on uncertain sensor data about the position and speed of nearby cars.

Techniques like Expected Utility Theory can be used to select an action that maximizes the expected benefit after evaluating the probabilities of different outcomes.

Applications of Probabilistic Reasoning in AI

- Medical Diagnosis
- Autonomous Systems
- Robotics
- Finance
- Speech and Language Processing

Advantages of Probabilistic Reasoning

- **Handles Uncertainty**

Can handle uncertainty, thus suitable for real-world applications where perfect information is rarely available.

- **Flexible and Scalable**

Can be applied to different application domains and can be used to model complex systems with many variables and dependencies.

- **Supports Inference**

Can help to make inferences based on partial information, thus can be used to draw conclusions even when not all data is available.

Limitations of Probabilistic Reasoning

- **Computational Complexity:** For large, complex systems with many variables, probabilistic reasoning can become computationally expensive due to the need to maintain and update probabilities for a large number of possibilities.
- **Requires Good Prior Knowledge:** The accuracy of probabilistic reasoning depends on having reliable prior probabilities, which may not always be easy to obtain.
- **Interpretability:** For some users, interpreting and understanding the results of probabilistic reasoning models (e.g., Bayesian Networks) can be challenging.